Differential Privacy against Membership Inference Attacks

A Case Study on Binary Classifiers

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AGENDA

Presentation Outline

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01 **Membership Inference Attacks**

Definition, Threat and Threat Model

02 **Target Model Implementation**

Vulnerable binary classifier on the Adult dataset

03 **Shadow Model Attack**

Attack architecture and pipeline

04 **Differential Privacy (DP-SGD)**

Mechanism, Opacus and privacy budget ε

05 **Iterative Tuning & Model Collapse**

Balancing issues and adopted solutions

06 **Results & Conclusions**

Utility-privacy trade-off and final metrics

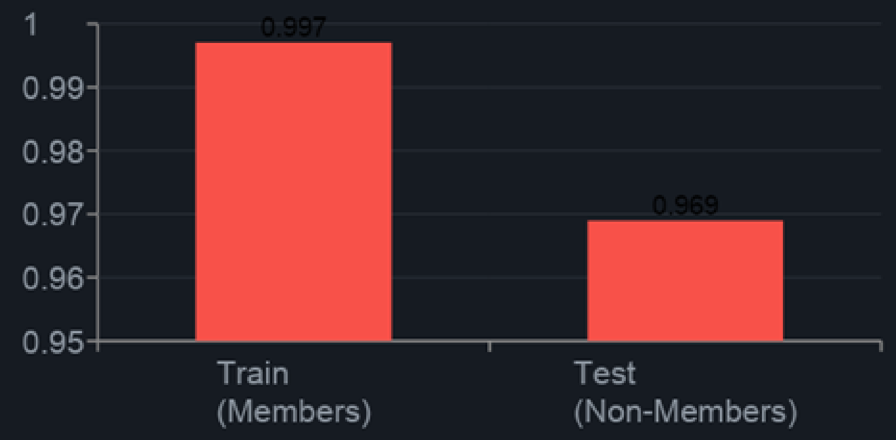
What is a Membership Inference Attack?

Definition: **An attacker**, given a data record x and access to the model (black-box or white-box), tries to determine whether *x ∈ D_train*.

Why is it dangerous?

- ML models may memorize patterns from the training data
- In MLaaS settings, the attacker can freely query the model
- Training records may reveal sensitive information, such as income or health data

Signal exploited by the attack



Adult Dataset & Vulnerable Target Model

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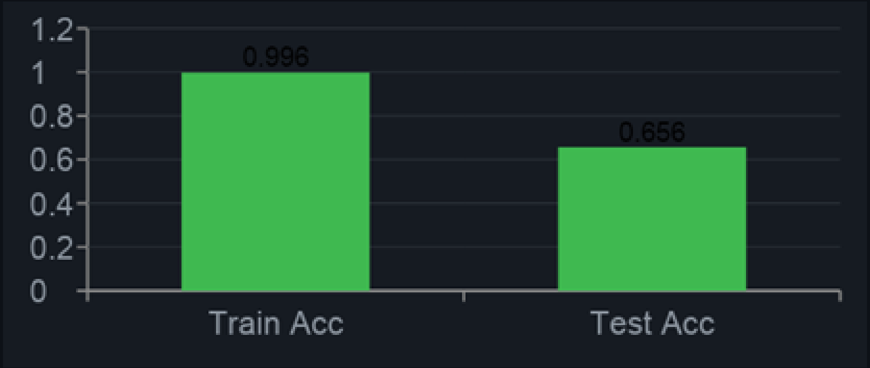
Adult Income Dataset

Task: **binary classification** (>50K / ≤50K)
Features: 14 attributes (age, occupation, education...)
Target split: 500 samples (250 train / 250 test)

Target MLP Architecture (epochs 200)

```
Linear(Input_Size, 256) → ReLU →
Linear(256, 128) → ReLU →
Linear(128, 64) → ReLU → Linear(64, 2)
```

Intentional Overfitting Gap



Baseline Metrics (no DP)

Train Accuracy	99.6%
Test Accuracy	65.6%
Overfitting Gap	34.0%

Attack Types

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Black-Box Score-Based

→ Access only to predictions

→ Observes confidence vectors

→Exploits the train/test gap

→ Used in this work

→ Approach: shadow models

FOCUS OF THIS WORK

Model-Agnostic (ML-Leaks)

→ Does not require an identical architecture

→ Uses only the score distribution

→ Salem et al. 2019

→ More generalizable

→ No overlap between dataset

FOCUS OF THIS WORK

White-Box Gradient

→ Access to weights and gradients

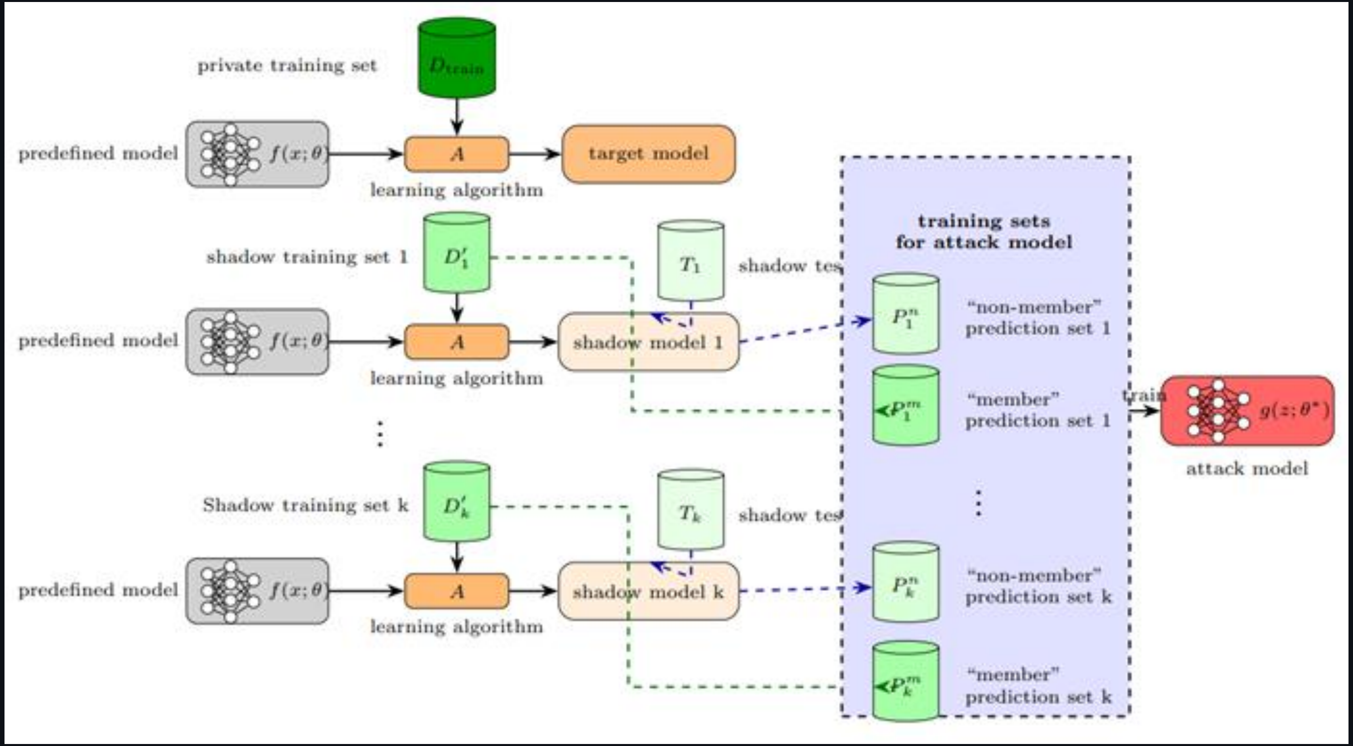
→ Uses influence functions

→ Leino & Fredrikson 2020

→ More powerful

→ Not covered in this study

Shadow Models



Attack Pipeline with Shadow Models

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1. Shadow Dataset (0% overlap)
2. 10 Trained Shadow Models
3. 'Member' vs 'Non-Member' Predictions
4. Attack MLP Training (binary clf)
5. Inference on the Target Model

Based on: Shokri et al. 2017 · ML-Leaks: Salem et al. 2019

Implementation Details

Component	Configuration
# Shadow Models	10 (with the same distribution as the target)
Shadow MLP	Different layers from the Target Model (ML-Leaks generalization)
Attack Classifier	Binary MLP: [In/Out] from the vector confidence
Baseline Result	Attack Accuracy 68.6% · Member Recall 1.00 · F1 0.76

Shadows & Attack model

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Shadows architecture (epochs 100)

```
Linear(Input_Size, 128) → ReLU →

Linear(128, 64) → ReLU → Linear(64, 2)
```

Attack model architecture (epochs 100)

```
Linear(Number_of_classes + 2, 64) → ReLU →

Dropout(0.3) →  Linear(64, 32) → ReLU →

Linear(32, 2)
```

```
# Max confidence
train_max_conf = train_probs.max(axis=1).reshape(-1, 1)
test_max_conf = test_probs.max(axis=1).reshape(-1, 1)
```

Vulnerability of the Unprotected Model

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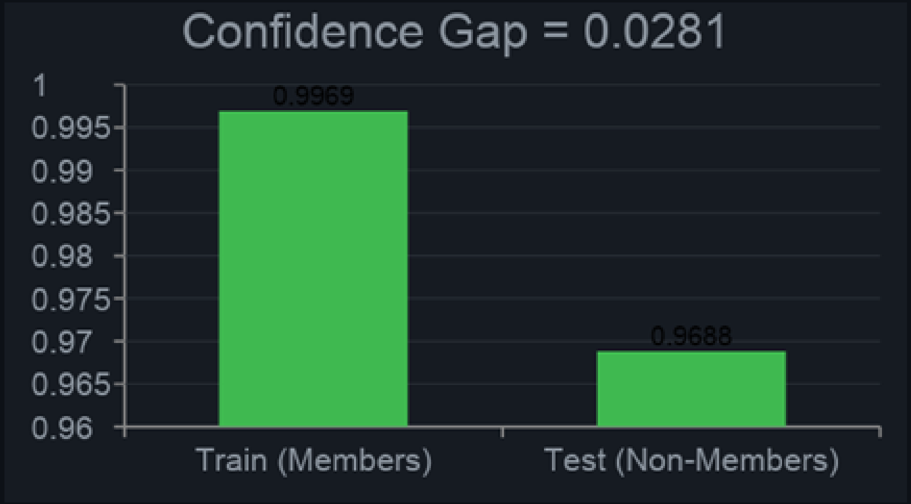
99.6%	65.6%	34.0%	68.6%
Train Accuracy	Test Accuracy	Overfitting Gap	Attack Accuracy



Attack Metrics

Attack Accuracy	68.6%
Member Precision	61.0%
Member Recall	100.0%
Member F1-Score	76.0%
Attack Advantage	+18.6%

Theoretical Foundations of DP-SGD

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Definition of (ε, δ)-DP: A mechanism M is (ε, δ)-differentially private if, for every pair of adjacent datasets D and D’ and every subset S of the output space: $Pr[M(D) \in S] \leq e^{\varepsilon} \cdot Pr[M(D') \in S] + \delta$

ε (Epsilon)

Privacy Budget

Measures the maximum privacy loss. Smaller values imply stronger privacy.

δ (Delta)

Failure Probability

Probability that the privacy bound fails. Dynamically set as: $\delta = 1/|X\_train| = 0.004$

σ (Sigma)

Noise Multiplier

Controls the intensity of the Gaussian noise injected into the gradients.

References: Abadi et al. 2016 – Deep Learning with Differential Privacy (CCS) · Humphries et al. 2020 – DP does not bound MIA in practice

Opacus: Meta’s Library for DP in PyTorch

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```
privacy_engine = PrivacyEngine()
model, optimizer, train_loader
  = privacy_engine.make_private(
    module=model,
    optimizer=optimizer,
    data_loader=train_loader,
    noise_multiplier=1.5,
    max_grad_norm=1.5
)
```

Privacy tracking in the code:

```
epsilon = privacy_engine.get_epsilon(delta)
```

Target threshold: ε < 10 · Final result: ε ≈ 5.31

1. **Per-Sample Gradients**
   Gradients computed for each individual sample, not immediately averaged

2. **Gradient Clipping**
   L2 norm clipped to max_grad_norm=1.5 to limit the influence of outliers

3. **Noise Addition**
   Calibrated Gaussian noise (σ=1.5) added to the clipped gradients

4. **Privacy Accounting**
   Opacus tracks the accumulated ε after epoch

Model Collapse: the trap of illusory privacy

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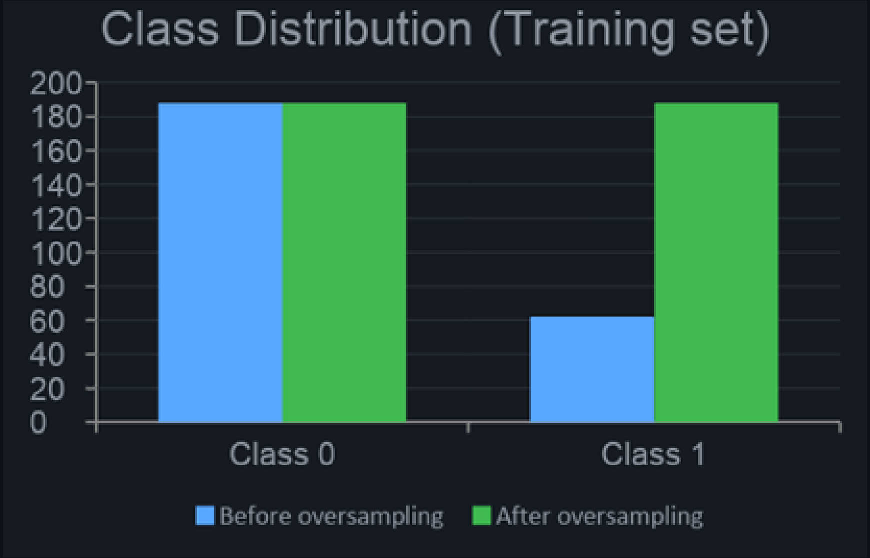
First DP attempt (clipping=1.0, σ=1.0, 200 epochs, batch=16): ε = 29.35, Test Accuracy ≈ 75.6%... but the confusion matrix showed **class 1 with F1-score of exactly 0.00.**



Adopted correction path

1 **Class-Weighted Loss**
Only partial improvement, sampling rate q≈0.256 was too high

2 **Oversampling**
Prevented collapse but duplicated data, with a risk of spurious overfitting

3 **Balanced Stratified Subset ✓**
250 samples per class, NO artificial duplication → final solution

Search for the Optimal DP Configuration

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Test	Batch	σ (Noise)	Epochs	Test Acc	Train Acc	Gap	ε
1 ★	32	1.5	40	69.6%	72.4%	2.8%	8.43
2	32	1.3	40	70.0%	72.0%	2.0%	10.54
3	32	1.5	30	67.2%	70.8%	3.6%	7.18
4	16	1.3	40	68.0%	68.8%	0.8%	6.98

Test 1 was the best result of the initial search, but not the final DP configuration. The final setup was obtained after further tuning: max_grad_norm = 1.5 and $\delta = 1 / |X\_train|$, instead of a fixed default δ value designed for larger datasets. This led to the final privacy budget $\varepsilon \approx 5.31$.

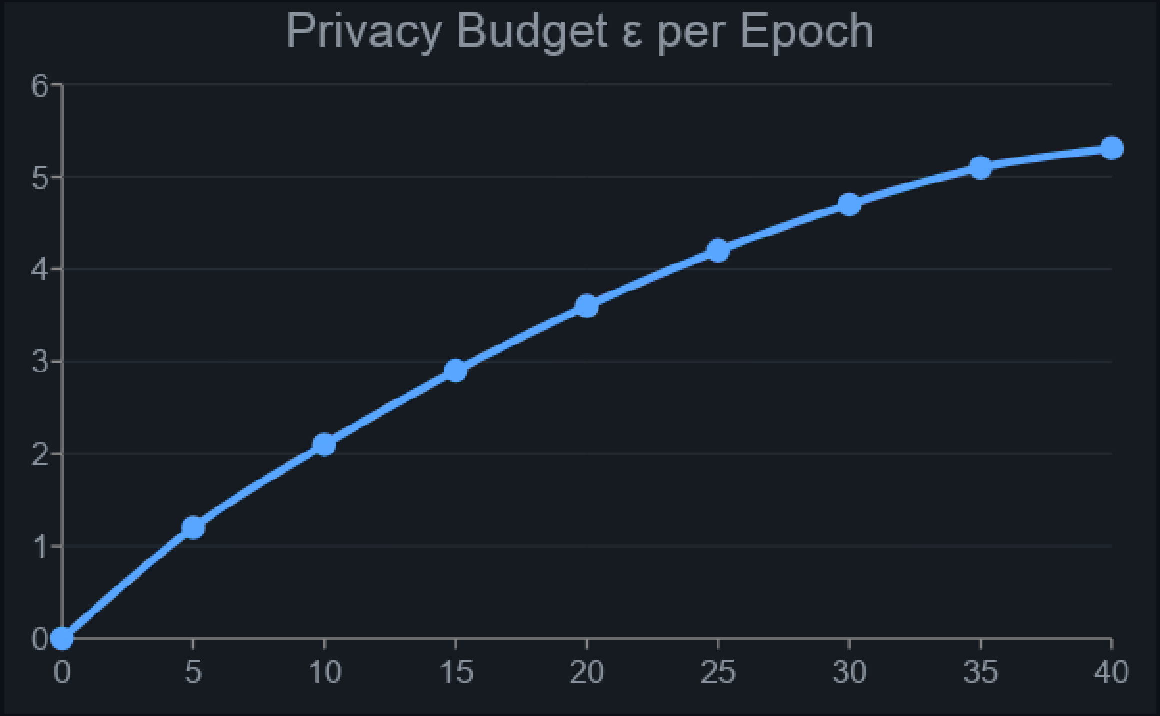
The Utility-Privacy trade-off

↓σ Reducing σ increases accuracy but increases ε (weaker privacy)

↓E Reducing the number of epochs limits ε accumulation but decreases accuracy

↓B Reducing the batch size improves privacy but makes training less stable

Evolution of ε during DP training



δ (failure prob.)

0.004 *1 / |X_train|*

σ (noise mult.)

1.5 *final configuration*

max_grad_norm

1.5 *clipping threshold*

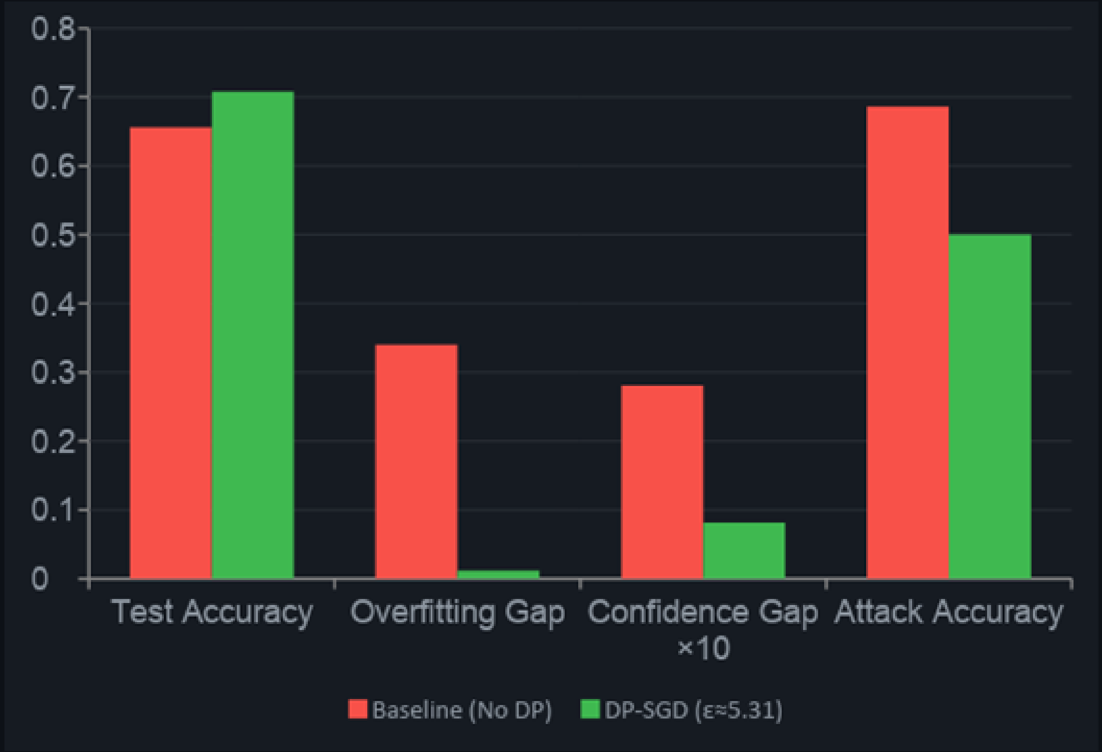
final ε (40 epochs)

5.31 *below the ε < 10 threshold*

Commonly accepted safety threshold: ε < 10

FINAL RESULTS

Comparison: Baseline vs DP-SGD



Overfitting Gap

34.0% → 1.2% ↓97%

Confidence Gap

0.0281 → 0.0082 ↓71%

Attack Accuracy

68.6% → ≈50% ↓random

Test Accuracy

65.6% → 70.8% (+5.2%)

Critical Analysis of Results and Limitations

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What Worked

✓ DP-SGD reduced the overfitting gap from 34.0% to 1.2%

✓ The confidence gap decreased from 0.0281 to 0.0082 (−71%)

✓ Attack accuracy dropped to ≈ 50% (close to the random baseline)

✓ Test accuracy even improved: 65.6% → 70.8%

✓ The balanced dataset avoided illusory privacy

✓ $\varepsilon \approx 5.31$ with $\delta = 0.004$: the formal privacy budget was respected

Limitations

⚠ Very small dataset (n=500): DP scales better on larger dataset

⚠ $\delta = 1/250$ accelerates budget depletion → more fragile tuning

⚠ Only black-box score-based attacks were covered; white-box attacks were not evaluated

⚠ Oversampling may introduce spurious patterns in privacy-sensitive contexts

⚠ Generalization to NLP or deep architectures requires further validation

Summary and Future Work

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DP-SGD (ε ≈ 5.31) effectively suppressed membership inference signals by reducing the overfitting gap to **1.2%** and bringing attack accuracy down to **≈ 50% (random baseline)**, while maintaining acceptable utility (Test Accuracy: **70.8%**). The key methodological correction was the use of a **perfectly balanced dataset (50/50)** to avoid illusory privacy.

Future Work

ROC-AUC and advanced metrics

Integration of ROC-AUC curves to evaluate inference boundaries

White-Box Analysis

Extension to white-box attacks with gradient access (Leino & Fredrikson)

Comparison with Regularization

Benchmark DP-SGD vs weight decay, dropout, early stopping

Larger dataset

Validate DP on real-scale tabular dataset (MLaaS setting)

Thank you for your attention

Any Questions?

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